Tue, Oct 27 at 3:54 AM

You said:

26.06.16 door sw…
File

These are message logs from a machine. the message format is generally as follows:
00:17:20:634 VM ROOT - LED 14 : state:4F --- B 08:00 - 08:8D 0E 01 4F
00:17:20:634 = timestamp
VM ROOT - LED 14 If a set of addresses are known, these are the labels for source - destination.
: state:4F if the message function is known, this is the label.
08:00 - 08:8D these are the source and destination addresses.
0E is a control field
01 indicates one data byte
4F is the data byte. Additional data bytes may follow.

Specifically, I am searching for any messages which might indicate that the machine door is opened or closed.
Many of the logs in the zip file contain multiple open/close actions on the door switch, as can be known by the log file.
Please review the files and make a list of plausible data packets which may contain the door switch data.
For example, the file "red machine door switch hold for 5 seconds 6 times.txt" would contain 6 events where the door was closed for 5 seconds, then opened (and then closed again.. 5 more times)
…

Show more

Worked for 31s

Analyzed

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ChatGPT said:

I'll look for packets that change in step with the known door-switch actions, especially in `red machine door switch hold for 5 seconds 6 times.txt`. The most useful candidates will be messages that repeat around each opening or closing, change their data bytes when the switch changes state, and appear less frequently in the logs without door activity.

One important distinction: a packet that changes when the door opens may report the door state directly, or it may report a related machine state that changes as a consequence. I'll treat these as plausible candidates rather than assume the meaning of any byte before the evidence supports it.

1. Strongest candidate: device-status packet

The clearest candidate I found is a packet sent from `VM ROOT` to the destination `08:80` in the red-machine test log.

Packet format:

```
D 08:00 - 08:80 0E 11 50 ...
```

The important observation is that the data bytes alternate between two distinct patterns during the test.

Observed data bytes	First observed timestamp	Occurrences	Interpretation
`50 DF FF FF AB FB BF FF FF FF FF FF FF FF FF FF FF`	`00:31:37:235`	6	One status pattern
`50 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF`	`00:31:41:645`	6	The alternate status pattern

These patterns recur throughout `red machine door switch hold for 5 seconds 6 times.txt`, with transitions occurring at approximately the intervals expected during the door-switch test.

Why this matters: The first data byte remains `50`, while several subsequent bytes change. This suggests that the packet may contain a collection of status flags, with one or more bits reflecting the switch state.

The available evidence does not yet establish which pattern means *door open* and which means *door closed*. However, the repeated alternation makes this packet a high-priority candidate for further investigation.

2. Second candidate: status packet with control field `0E 0A`

Another promising packet appears in several of the other logs, including both files explicitly named for opening and closing the door.

Its address and control fields are:

```
D 08:00 - 08:80 0E 0A
```

I found three related data patterns:

Data bytes	Example timestamp	Observation
`D1 00 00 00 00 00 00 00 00 00`	`00:17:22:647`	Appears as part of a three-packet sequence
`50 FF FF FF FF FF FF FF FF FF`	`00:17:22:667`	Appears immediately afterward
`58 55 55 55 FF FF FF F5 FF FF`	`00:17:22:687`	Appears immediately afterward

The same three-packet sequence occurs in `door open door close 2.txt` at approximately `00:18:52.791–00:18:52.825`.

It also appears repeatedly in `white machine 5 minutes on the head.txt`, approximately every 30 seconds.

In `white terminal lockout price button fails open door close door then change prices.txt`, I found a related pattern:

```
58 FF FF FF FF FF FF FF FF FF
```

This appears at `00:33:25:910`, `00:33:42:484`, and `00:33:51:030`, interspersed with occurrences of the `58 55 55 55 FF FF FF F5 FF FF` pattern.

Assessment: These packets are worth investigating, particularly because their data bytes change while the destination remains the same. However, their timing and recurrence suggest that at least some may be associated with periodic status polling or other machine activity rather than exclusively with the door switch.

I would rank this family below the `0E 11` packet for now.

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3. Other packet families worth investigating

The logs also contain other messages that could be relevant, but the evidence is weaker.

Priority	Packet family	Why investigate it?
1 — High	`D 08:00 - 08:80`, control `0E 11`, data beginning with `50`	Alternates between two distinct payloads during the six-cycle door-switch test.
2 — Medium	`D 08:00 - 08:80`, control `0E 0A`, data beginning with `D1`, `50`, or `58`	Occurs in several door-related logs, with different payloads.
3 — Exploratory	`VM ROOT - --------`, destinations `0B:80`, `0B:81`, etc., control `0E 11`	Contains changing status-like bit patterns, but the relationship to door activity is not established.
4 — Exploratory	`VM ROOT - HEAD`, destination `09:01`, data beginning with `E3`, `E5`, or `E1`	Some values change during the door-related logs, but they may reflect head or machine operation instead.

The third and fourth families should not be treated as door-switch signals without further comparison against known event times.

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4. What I would investigate next

The most useful next step is to compare the six known opening and closing cycles against the exact timestamps of every packet in the red-machine test.

For each cycle, I would identify:

1. The last occurrence of each candidate packet before the switch changes.
2. The first occurrence after the change.
3. Whether the same data-bit transition occurs in all six cycles.
4. Whether the same transition is absent from logs where the door remains closed.
5. Whether the corresponding transition reverses when the door closes.

The `0E 11` packet is particularly promising because its two observed payloads differ in a repeatable way. If the same bit or bits change at every known opening and reverse at every known closing, that would be much stronger evidence of a door-state indicator.

My conclusion

Start with `D 08:00 - 08:80`, control field `0E 11`, and compare these two payloads:

- `50 DF FF FF AB FB BF FF FF FF FF FF FF FF FF FF FF`
- `50 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF`

Then investigate the `0E 0A` packets as a secondary candidate.

I would not assign a definitive open/closed meaning to either pattern yet. The next comparison should establish whether the transitions consistently match the known physical switch events.

I can also produce a **timestamped event-correlation table for all six cycles**, showing the packets immediately before and after each suspected opening and closing.

You said:

one helpful bound may be that the keyboard display text is redrawn, in messages marked VM ROOT - KEYBOARD : Row 1 to Row 4

ChatGPT said:

That gives us a useful way to distinguish **door-switch events from keyboard-display activity**.

In the red-machine test, the four keyboard-row messages are sent within roughly 100 milliseconds of one another. For example, at `00:31:41`, the rows are redrawn from

`00:31:41:580` through `00:31:41:686`, with the candidate `08:80 / 0E 11` packet appearing at `00:31:41:645`—in the middle of the redraw.

This means we should be careful not to interpret every packet near a display redraw as a door event.

I'd use the keyboard messages as follows:

- **Identify redraw windows:** Group the `VM ROOT - KEYBOARD : Row 1` through `Row 4` messages into display-redraw events.
- **Compare candidate packets:** Determine whether the candidate packet changes consistently at the known door-switch transitions, including transitions that occur outside redraw windows.
- **Separate display effects:** If a packet changes only when the keyboard display redraws, it may be associated with display activity rather than the door switch.
- **Look for independent evidence:** Prioritize packets whose payloads change at the known door-switch transitions even when no keyboard-row redraw occurs.

One caveat: a redraw could itself be triggered by the door opening, so I wouldn't automatically discard packets that coincide with one.

The key question is whether the `08:80 / 0E 11` packet tracks the physical switch independently of the keyboard redraws. Comparing all six cycles against the redraw timestamps should help establish that.

You said:

long idle.txt
Document

let me give you another log. In this log, the door is simply OPEN for nearly 5 minutes. nothing of note is happening with the machine, so we refer to it as "idle".
If any of the above discovered patterns appear here, you can eliminate them.

Worked for 30s

ChatGPT said:

The idle log is a valuable negative control: since the door remains open for nearly five minutes without significant machine activity, a packet that appears repeatedly in this log is unlikely to be a reliable indicator of a *door opening or closing event*.

There is an important distinction, though: **a packet that occurs while the door is open is not necessarily irrelevant.** It could report the door's current state rather than a transition. I'll distinguish packets that can be eliminated as transition indicators from those that might still report a persistent open-door state.

Results from `long idle.txt`

I checked the 3,546-line idle log against the candidate packets identified in the earlier logs.

Candidate	Found in idle log?	Updated assessment
`08:00 → 08:80`, control `0E 11`, payload beginning `50 DF FF FF AB FB BF`	No	**Retain as a strong candidate.**
`08:00 → 08:80`, control `0E 11`, payload beginning `50 FF FF FF FF FF FF`	No	**Retain as a strong candidate.**
`08:00 → 08:80`, control `0E 0A`, payload beginning `D1 00 00 00`	No	Retain for further investigation.
`08:00 → 08:80`, control `0E 0A`, payload beginning `50 FF FF FF`	No	Retain for further investigation.
`08:00 → 08:80`, control `0E 0A`, payload beginning `58 55 55 55`	No	Retain for further investigation.
`08:00 → 0B:xx`, control `0E 11`	Yes	Deprioritize as a door-transition indicator.

An important observation

The `08:00 → 0B:xx` packets appear regularly throughout the idle log. For example, the destination addresses `0B:80` through `0B:9D` recur in a repeating sequence, approximately once per second.

This regularity suggests that these packets belong to a periodic polling or status-reporting process. They are therefore much less compelling as indicators of individual door-opening and door-closing events.

However, I would not rule out the possibility that their payloads contain a persistent door-state bit without examining the relevant bytes across the test logs.

Updated conclusion

None of the five specific `08:80` payload patterns identified above appears in the idle log. This is encouraging: the patterns remain plausible candidates for door-related messages.

The next step is to compare the five candidates against the six known door-switch cycles and determine which payload changes align most closely with the physical switch transitions.

One qualification: the absence of these packets during this particular idle recording does not prove they indicate the door. It does, however, provide useful evidence for prioritizing them in the next round of analysis.

You said:

boot and 5 minut…
Document

boot and 5 minut…
Document

here are two more 5 minute long idle logs, this time on a different machine. They may help to refine the eliminations.

Worked for 27s >

Analyzed

Analyzed

Analyzed

Analyzed

ChatGPT said:

These two new logs substantially strengthen one of the eliminations: **the `08:00 → 08:80` messages with control field `0E 0A` are not reliable door-transition indicators by themselves.**

Both recordings contain these messages repeatedly while the machines are idle. For example, the first log shows `50 FF FF...`, followed by `58 00 00...`, and then another `50 FF FF...` message; the second log shows the same general pattern.

boot and 5 minute idle | boot and 5 minute idle 2

Updated elimination results

Candidate	Evidence from the two new logs	Assessment
`08:00 → 08:80`, `0E 0A`, payload beginning `50 FF FF...`	Appears **90 times** in the first log and **87 times** in the second.	**Eliminate as a unique door-transition indicator.**
`08:00 → 08:80`, `0E 0A`, payload beginning `58 55 55...`	This exact pattern does not appear in either new log, although other `58` payloads do.	**Keep as an unconfirmed candidate**, but downgrade confidence in the overall message family.
`08:00 → 08:80`, `0E 0A`, payload beginning `D1 00...`	Does not appear in either new log.	**Still unresolved**; these recordings don't rule out this exact payload.
`08:00 → 08:80`, `0E 11`, payload beginning `50 DF...` or `50 FF...`	Neither candidate appears in these logs. The `08:80` traffic here uses other control fields, chiefly `0E 0A`.	**Retain for further comparison** with the known door-switch test.

What the new logs tell us

Each recording contains more than **1,000 messages** addressed to `08:80` with control field `0E 0A`—1,056 in the first and 1,044 in the second. This strongly suggests that this message family includes routine traffic, not just door events.

One important distinction: a message family can be routine while a *particular payload* still reflects a meaningful state. So I would eliminate the recurring `50 FF FF...` payload as a unique door-transition signal, but not automatically discard every `0E 0A` payload.

Current priority

1. **Prioritize the `0E 11` candidates** that appeared in the six-cycle door-switch test but are absent from these new logs.
2. Keep the exact `58 55 55...` and `D1 00...` payloads as lower-confidence candidates until we compare them against the known door transitions.
3. Deprioritize the recurring `0E 0A` `50 FF...` pattern.

These are eliminations based on the available recordings, not proof of the protocol's meaning. The next decisive step remains aligning candidate packets with the exact

timestamps of the six physical door transitions.

You said:

could you print out exact samples from the logs of the messages to check?

Worked for 16s

ChatGPT said:

Yes. Here are **exact lines copied from your logs**, with timestamps and line numbers, so you can compare them against what the door was physically doing.

1. Main candidates from the six door-switch cycles

File: `red machine door switch hold for 5 seconds 6 times.txt`

These two message patterns each appear six times in the test where you operated the door six times.

Pattern A — payload starts `50 DF`

Plain text

```
Line 37:  00:31:37:235  VM ROOT  - -------- : --------  --- D 08:00 - 08:80  0E 11 50 DF
Line 230: 00:31:47:742 VM ROOT  - -------- : --------  --- D 08:00 - 08:80  0E 11 50 DF F
Line 400: 00:31:56:828 VM ROOT  - -------- : --------  --- D 08:00 - 08:80  0E 11 50 DF F
Line 590: 00:32:08:734 VM ROOT  - -------- : --------  --- D 08:00 - 08:80  0E 11 50 DF F
Line 763: 00:32:16:839 VM ROOT  - -------- : --------  --- D 08:00 - 08:80  0E 11 50 DF F
Line 955: 00:32:27:451 VM ROOT  - -------- : --------  --- D 08:00 - 08:80  0E 11 50 DF F
```

Pattern B — payload starts `50 FF`

Plain text

```
Line 131:  00:31:41:645 VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 11 50 FF FF
Line 330:  00:31:51:746 VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 11 50 FF FF
Line 477:  00:32:01:647 VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 11 50 FF FF
Line 704:  00:32:13:430 VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 11 50 FF FF
Line 880:  00:32:22:736 VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 11 50 FF FF
Line 1049: 00:32:31:641 VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 11 50 FF FF
```

These are the strongest initial candidates because both patterns repeat six times in the controlled door test. We still need the exact physical open/close timing to determine which pattern means which state.

For context, here is one of the surrounding sequences from the same file:

```
Plain text

00:31:41:610  VM ROOT - KEYBOARD : Row 2     --- B 08:00 - 09:31 0E 13 00 17 16 20 20 20
00:31:41:640  VM ROOT - VEND NEW : --------  --- B 08:00 - 0E:C0 0E 05 30 7F 06 03 00
00:31:41:645  VM ROOT - -------- : --------  --- D 08:00 - 08:80 0E 11 50 FF FF FF FF FF
00:31:41:654  VM ROOT - KEYBOARD : Row 3     --- B 08:00 - 09:31 0E 13 00 33 16 BA B3 D3
00:31:41:663  VEND NEW - VM ROOT : --------  --- B 0E:C0 - 08:00 0E 07 B0 00 02 7F 06 03
```

Notice how the candidate appears among other keyboard and vending messages. That means we should correlate it with the exact door action, rather than assuming the nearby keyboard messages are the door signal.

2. The `0E 0A` messages from the door-open/door-close logs

These are also worth checking, but the new idle recordings show that some of this family occurs during idle operation.

Pattern C — `D1 00 ...`

From `door open door close.txt`, line 31:

```
Plain text

00:17:22:647  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A D1 00 00 00 00 00 0
```

From `door open door close 2.txt`, line 138:

```
Plain text

00:18:52:791  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A D1 00 00 00 00 00 0
```

This exact payload also appears every 30 seconds in `white machine 5 minutes on the head.txt`, for example:

```
Plain text

00:01:59:619  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A D1 00 00 00 00 00 0
00:02:29:679  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A D1 00 00 00 00 00 0
00:02:59:758  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A D1 00 00 00 00 00 0
```

That regular interval suggests it may be periodic traffic, not necessarily a door transition.

Pattern D — `50 FF ...`

From `door open door close.txt`, line 32:

```
00:17:22:667  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A 50 FF FF FF FF FF F
```

From the first new idle recording, line 171:

```
00:02:00:783  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A 50 FF FF FF FF FF F
```

And from the second new idle recording, line 164:

```
00:15:02:783  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A 50 FF FF FF FF FF F
```

This is **not a unique door-event marker**: it appears in recordings where the machine was left idle.

Pattern E — `58 55 55 ...`

From `door open door close.txt`, line 33:

```
00:17:22:687  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A 58 55 55 55 FF FF F
```

From `door open door close 2.txt`, line 140:

```
00:18:52:825  VM ROOT - -------- : -------- --- D 08:00 - 08:80 0E 0A 58 55 55 55 FF FF F
```

However, the same exact message appears in the five-minute recording in `white machine 5 minutes on the head.txt`, repeatedly at 30-second intervals. So this exact payload also needs more scrutiny before we can label it a door signal.

3. What I would check first

Candidate	What the samples show	Current assessment
`0E 11 50 DF FF FF AB FB BF...`	Appears six times in the six-cycle door test	**High-priority candidate**
`0E 11 50 FF FF FF FF...`	Appears six times in the same test, alternating with Pattern A	**High-priority candidate**
`0E 0A D1 00 00...`	Also repeats periodically in another recording	Probably routine or periodic traffic; unconfirmed
`0E 0A 50 FF FF...`	Appears during idle recordings	Not a unique door indicator
`0E 0A 58 55 55...`	Appears repeatedly in a five-minute recording	Not a unique door indicator by itself

The most useful next step is to compare Patterns A and B with your actual open/close sequence. If you can tell me the approximate timestamps when you opened and closed the door in the six-cycle test—or provide the timing notes for that test—I can align each packet to the physical action and try to identify which one corresponds to opening versus closing.